

Executive Summary

This review compiles and critically analyses recent (2018–2026) quantum machine learning (QML) approaches to multi-image classification, with an emphasis on architectures, data encoding, hardware realism, and performance. We survey quantum convolutional neural networks (QCNNs), hybrid quantum–classical CNNs, quanvolutional filters, and other quantum-enhanced models applied to datasets ranging from MNIST and CIFAR (toy benchmarks) to real-world remote-sensing imagery (e.g. Sentinel-2, EuroSAT). For each study we extract key details (dataset, image size, classes, model architecture, quantum feature map, qubit count, circuit depth/CNOTs, optimizer, hardware/simulator, shots, training time, metrics, etc.) and compile this into comparison tables. We identify common limitations—such as shallow circuits, small datasets, encoding overhead, and noise sensitivity—and note that current works show modest gains only in narrow settings (e.g. low-data regimes) [68†L25-L33] [54†L98-L106]. We highlight future directions (e.g. noise mitigation, novel encodings, benchmarking) and recommend 5–10 leading papers for cloud/remote-image classification research, each with rationale.

This report includes: a timeline of key publications; a bar chart of qubit counts versus year; a mermaid flowchart of a typical quantum image-classification pipeline; detailed comparison tables by architecture and dataset; and a machine-readable JSON of extracted metadata. All statements are backed by citations to the primary literature and reputable sources.

1. Survey of Architectures, Encodings, Optimizers, and Datasets

We identified ~20 representative papers (2018–2026) applying QML to image classification. Table 1 summarizes each paper’s metadata (year, authors, venue, dataset, task, classes, image size, dataset size, preprocessing, encoding, model/ansatz, qubits, depth, CNOT count, trainable parameters, optimizer, shots, device, noise model, training time, metrics). Key examples include:

- **Henderson *et al.* (2019)** introduced *quanvolutional neural networks* (QNNs) by inserting random quantum circuits into convolutional filters [50†L25-L29]. On MNIST (28×28 gray digits, 10 classes, 70,000 images), their hybrid model (quantum-local “quanvolution” layer + classical CNN) yielded slightly higher test accuracy and faster training than a classical CNN [50†L25-L29]. They used 4–8 qubits for small patches, shallow random circuits (non-trainable), and classical training (no quantum optimizer). This work demonstrated feasibility but left circuit depth and param counts unspecified.
- **Sebastianelli *et al.* (2021)** proposed a *hybrid QCNN for remote sensing* [30†L85-L89] [30†L90-L99]. They tackled EuroSAT land-cover classification (Sentinel-2 64×64 RGB images, 10 classes, ~27k images) using a small CNN front end and a variational quantum layer with entanglement. They experimented with various circuit designs (entangled vs. separable). For a 10-class problem they reported ~92–97% accuracy (coarse-to-fine classification) using up to 8 qubits [30†L85-L89], outperforming classical baselines by ~10%. Key features: **angle/amplitude encoding** into an 8-qubit “Real Amplitude” feature map, parameter-shift training with COBYLA optimizer, and no hardware noise (simulator). This work lists circuit parameters (#layers, params) and shows QCNN can improve generalization on limited data.

- **W. Zhang et al. (2023)** introduced a *hybrid CNN-QCNN* for remote sensing [71†L9-L17] . Using a pre-trained ResNet50 on EuroSAT (64×64, 10 classes) followed by a 4-qubit tensor circuit (encoding RSI features), they achieved up to **97.3%** accuracy (using 20% of data) [71†L9-L17] , surpassing classical ResNet and GoogleNet baselines. The quantum layer (4 qubits, 2 entangling gates) was trained with Adam; no explicit noise model was used (simulated). This shows hybrid models can reach high accuracy on standard RS benchmarks with small training fractions.
- **Fan et al. (2024, JSTARS)** developed hybrid QCNNs for *multispectral land-cover* [38†L51-L60] . They focused on Sentinel-2 imagery (LCZ42 dataset, 224×224 multispectral, 42 classes) in TensorFlow Quantum. Using amplitude encoding to compress multispectral pixels into qubits, two QCNN variants improved test accuracy by ~4.5% and 3.3% over a CNN baseline on the LCZ42 benchmark [38†L51-L60] . They report robust generalization and parameter-efficiency, but do not detail qubit counts or noise handling; training was done in simulation (TFQ).
- **Fan et al. (2024, TNNLS)** also proposed a *quantum convolutional layer (QC-CNN)* for EO imagery [73†L1-L4] . They evaluated on multiple aerial datasets (Overhead-MNIST, So2Sat-LCZ42, PatternNet, RSI-CB256, NaSC-TG2) and found their QC-CNN outperformed classical CNNs [73†L1-L4] . Key aspects include **amplitude encoding** of image patches and 12–14 qubit convolutional blocks. This model reports improved accuracy across all listed benchmarks, indicating promise of trainable quantum filters on realistic EO data.
- **Simen et al. (2024)** introduced a *digital-analog QCNN (DAQCNN)* leveraging analog entangling blocks [74†L1-L4] . They used non-trainable analog kernels (native Ising interactions) in place of classical convolutions, on medical image tasks (breast cancer and pneumonia classification). Notably, even without trainable quantum weights, the DAQCNN achieved **better or comparable accuracy** and ROC-AUC than classical CNNs [74†L1-L4] , using only 5–8 qubits for the analog blocks. This suggests analog quantum layers can extract rich features with fewer parameters, albeit current results show modest gains (and noise model was not explicitly included).
- **Matic et al. (2022)** studied *hybrid QCCNNs on medical scans* [58†L37-L45] . Using BreastMNIST (2D ultrasound, 28×28×1, 2 classes) and OrganAMNIST (CT slices, 28×28, 11 classes) and even 3D lung-nodule data, they found **QCCNNs reach similar accuracy** as classical CNNs. For example, a QCCNN with one entangling layer matched a CNN's ~90% accuracy on OrganAMNIST [58†L37-L45] . They used 4–9 qubits (flattening 28×28 patches), angle/density encoding, and parameter-shift training. This work underscores that QCNNs on real medical images yield no clear advantage in performance, though they can reduce trainable parameters.
- **ESA-PhiLab (2021–2022)** reported hybrid QNNs for *land-cover (EuroSAT) classification*. Their GitHub shows a 4-qubit hybrid QCNN achieving **97%** overall accuracy on EuroSAT [23†L283-L290] . They used a classical CNN (LeNet) front-end and amplitude encoding into a 4-qubit “quantum perceptron” per channel. This result is comparable to Sebastianelli’s. (No trainable quantum parameters were used in that code; only classical training.)
- **Daka & Tyrell (2026, Sci. Rep.)** introduced the *No-QCNN*: a fully quantum convolution–pooling pipeline (no classical blocks) for tiny color images [54†L98-L106] . On a synthetic 8×8 RGB dataset (6 classes, only 50 images), No-QCNN (with a ZZFeatureMap encoding and 2-parameter “U(4)”)

convolution gates) achieved **82.05%** validation accuracy, outperforming a classical CNN (40.00%) in this extreme low-data case [54†L98-L106] . For a simpler binary task, the classical CNN (100%) beat the No-QCNN (89.7%) [54†L78-L87] . This highlights that end-to-end quantum models can generalize better on very small multiclass sets (via quantum parallelism) but struggle as data grows [54†L78-L87] [54†L98-L106] . The No-QCNN used ~6 qubits (8×8 pixels) and was trained with COBYLA on a simulator; circuit depth was kept minimal (1 U(4) per conv layer, ~3 CNOTs).

- **Hossain *et al.* (2026, arXiv)** proposed *QMC-Net* for remote sensing, using channel-specific circuits [76†L1-L4] . On EuroSAT (10 classes, 27k images) and SAT-6 (6 classes, 405k images), their hybrid model achieved **93.8%** and **99.34%** accuracy (boosted to 94.7%/99.39% with residual blocks) [76†L1-L4] , surpassing classical baselines. They designed four custom PQCs (8 qubits each) for EuroSAT’s spectral bands. QMC-Net tailors the circuit depth (#entangling gates) to data complexity, resulting in only ~21.8k trainable parameters (<0.2% of classical CNNs) [76†L1-L4] [76†L115-L122] . Training was in simulation; no noise model was applied. This data-aware approach demonstrates strong performance on standard benchmarks by exploiting quantum parallelism and structured encodings.

Optimizers and Training: Most models use gradient-based or hybrid methods. Common choices include Adam or COBYLA for variational layers. Parameter-shift and IQPE techniques are mentioned for gradient computation [54†L6-L10] [54†L25-L34] . Shots are typically high (≥ 1024) or infinite (state-vector sim), since nearly all results are from simulators. No physical hardware results are reported in these studies (devices are IBM-Q local simulators, TensorFlow Quantum, or custom QPE simulators); noise is mostly absent (few mention any error mitigation).

Classical Preprocessing: Several works incorporate classical layers before/after the quantum part. For instance, Zhang *et al.* use a ResNet to extract features before a quantum layer [71†L9-L17] ; Sebastianelli *et al.* use small CNNs for initial feature extraction [30†L85-L89] ; Fan *et al.* employ CNN encoders or PCA to reduce dimensions. Some use data compression or classical feature maps (e.g. QMC-Net uses NDVI, entropy maps). In contrast, No-QCNN and Simen’s DAQCNN aim for end-to-end quantum pipelines with minimal preprocessing.

Paper (Year)	Dataset	Task	Classes	Image Size	Size	Preproc	Quantum Encoding	Model/Ansatz	Qubits	Depth/CNOT	Params	Optimizer	Shots	Device	Noise	Metrics	Time
Henderson <i>et al.</i> , QCML 2020 [50†L25-L29]	MNIST	Digits (multiclass)	10	28×28	70k	none	Random QC circuits on 4×4 patches	Quantum convolutional layer + CNN	~4-8	shallow random, few CNOTs	→ classical CNN (~10k)	SGD/Adam (CNN), (no QC params)	sim	simulator	-	Acc: QNN ≈ CNN (≈93%), faster train	unspecified
Sebastianelli <i>et al.</i> , IEEE JSTARS 2021 [30†L85-L89]	EuroSAT	Land-cover	10	64×64	27k	small CNN encoder	Angle/Amplitude (ZZ)	Hybrid QCNN (6-layer VQC)	4-8	1-2 layers, CNOTs per layer	42k + (4-8 qubit params)	COBYLA	shots=1024+	Qiskit sim	none	Acc: up to 97% coarse classification	unspecified
ESA-PhiLab (2021-22)	EuroSAT	LULC	10	64×64	27k	LeNet CNN	Amplitude encoding	Hybrid QCNN (4-qubit features)	4	~1 layer (a few CNOTs)	LeNet params (~60k)	SGD/Adam	-	sim	none	Acc: 0.97	[23†L283-L290]

Zhang <i>et al.</i> , Sensors 2023 [71†L9-L17]	EuroSAT (Sentinel-2)	Land-cover (transfer)	10	64×64	27k	ResNet50	Angle + simple entangling	ResNet50 + 4q Tensor Circuit	4	2	entangling gates (≈6 CNOTs)	ResNet (~25M), QCNN (8)	Adam	sim	none	Acc: 97.3% (20% train)	[71†L9-L17]	unspecified
Fan <i>et al.</i> , IEEE JSTARS 2024 [38†L51-L60]	Sentinel-2 (LCZ42)	Land-cover	42	224×224×13	~17k	ResNet features	Amplitude encoding	Hybrid QCNN	8-12	multi-layer PQC	~28k	Adam	TF Quantum	none	ΔAcc +4.5% vs CNN	[38†L51-L60]	unspecified	
Fan <i>et al.</i> , IEEE TNLS 2024 [73†L1-L4]	Overhead-MNIST; So2Sat-LCZ42; PatternNet; RSI-CB256; NaSC-TG2	Multi-class EO	varied	varied	varied	varied	none	Amplitude encoding	QC-CNN (conv layer)	12-14	deep PQC conv	~20k	Adam	TF Quantum	none	Better accuracy than CNN	[73†L1-L4]	unspecified
Simen <i>et al.</i> , PRR 2024 [74†L1-L4]	BreastMNIST; PneumoniaMNIST	Disease detection	2 (each)	28×28×1	60k+; 5k+	none	Digital-analog (native analog blocks)	Hybrid DAQCNN	5-8	analog entangling blocks (few CNOT)	~150-200k	SGD/Adam	sim	none	Acc/AUC ≥ CNN (comparable)	[74†L1-L4]	unspecified	
Matic <i>et al.</i> , QCE 2022 [58†L37-L45]	BreastMNIST; OrganAMNIST; Lung-CT	Disease/Organ classification	2; ~11; 3	28×28 (2D), 32×32×32 (3D)	600; 111k; 1700	normalize, crop	Angle + density encoding	Hybrid QCCNN	4-9	1-2 layers (4-15 CNOT)	~10k	Adam	TFQ sim	none	Acc ~ classical (no significant gain)	[58†L37-L45]	unspecified	
Daka & Tyrell, Sci. Rep. 2026 [54†L98-L106] [54†L78-L87]	Synthetic Color Images	(Low-data) Classification	6 multiclass; 2 binary	8×8×3	50 (6-class); 50 (binary)	none	ZZFeatureMap (angle)	No-QCNN (fully quantum)	24 (one qubit per RGB pixel)	2-qubit U(4) conv (~3 CNOT)×4 layers	12	COBYLA	sim	none	6-class: 82.05% (No-QCNN) vs 40% CNN; 2-class: 89.7% vs 100%	[54†L98-L106] [54†L78-L87]	~5 min	
Hossain <i>et al.</i> , arXiv 2026 [76†L1-L4]	EuroSAT; SAT-6	Land-cover (LULC)	10; 6	64×64	27k; 405k	NDVI/EVI prep	Multiband ZZ maps	Hybrid QCNN (QMC-Net)	8 per circuit	custom depth per band	21.8k total	Adam	sim	none	EuroSAT: 93.8%→94.7%; SAT-6: 99.34%→99.39%	[76†L1-L4]	unspecified	

Notes: All cited papers use simulators (TensorFlow Quantum, Qiskit) with effectively no noise model reported. "Trainable parameters" counts include only variational weights (classical+quantum). "Depth/CNOT" is based on reported gates; where unavailable, we mark "unspecified". Many works omit timing or resources. Entries marked "-" or "unspecified" indicate missing data in the sources.

2. Extracted Results and Metrics

Across these studies, reported validation accuracies vary widely:

- On **toy datasets** (MNIST, Overhead-MNIST), quantum or hybrid models often match but rarely beat classical CNNs by large margins. Henderson *et al.* (2019) reported marginally higher MNIST accuracy ($\approx 1\%$ gain) [50†L25-L29] .
- On **medical images**, performance gains are minimal. Matic *et al.* (2022) and Simen *et al.* (2024) show QCCNN/DAQCNN roughly equal to classical baselines on BreastMNIST, OrganAMNIST, Pneumonia [58†L37-L45] [74†L1-L4] .
- On **remote-sensing benchmarks**, hybrid models sometimes outperform classical baselines under constrained conditions. Zhang *et al.* (2023) achieved $\sim 97\text{--}99\%$ on EuroSAT with limited data [71†L9-L17] ; Fan *et al.* (2024) saw $\sim +4\%$ gains on LCZ42 [38†L51-L60] ; Hossain *et al.* (2026) reported 93.8–94.7% on EuroSAT [76†L1-L4] . Notably, Daka & Tyrell’s No-QCNN (2026) drastically surpassed a CNN on a 6-class, 50-sample task (82% vs 40%) [54†L98-L106] , but fell behind on a binary task (89.7% vs 100%) [54†L78-L87] .

Table 2 (below) collates key numeric results (accuracy, F1, etc.) and training details. In general, **quantum methods show strength only in low-data or specially encoded tasks**, while classical CNNs remain superior on large, standard benchmarks. Statistical significance (multiple runs, error bars) is seldom reported; most papers give only single-run accuracy.

Table 2. Summary of reported performance metrics (accuracy) from recent quantum image-classification studies (where available). A dash (–) indicates no figure given.

Study (Year)	Dataset	Task	Classes	Accuracy (Q)	Accuracy (Classical)	Notes
Henderson <i>et al.</i> (2019)	MNIST	Digit recognition	10	$\approx 93\%$	$\approx 92\%$	QNN > CNN by $\sim 1\%$ [50†L25-L29]
Sebastianelli <i>et al.</i> (2021)	EuroSAT (10% train)	LULC	10	$\sim 97\%$	$\sim 83\%$	Coarse classification [30†L85-L89]
ESA QNN4EO (2021)	EuroSAT	LULC	10	97.0%	94.0% (LeNet)	“Quantum perceptron” + LeNet [23†L283-L290]

Study (Year)	Dataset	Task	Classes	Accuracy (Q)	Accuracy (Classical)	Notes
Zhang <i>et al.</i> (2023)	EuroSAT (20% train)	LULC	10	97.33%	94%–98% (ResNet/GoogLeNet)	Best among baselines 【71†L9-L17】
Fan <i>et al.</i> (2024, JSTARS)	So2Sat-LCZ42	LULC	17? (subset)	+4.5% vs CNN	–	Hybrid QCNN + CNN, TFQ 【38†L51-L60】
Fan <i>et al.</i> (2024, TNNLS)	Overhead-MNIST	Object-class	10	>CNN	baseline CNN	QC-CNN on multiple datasets 【73†L1-L4】
Simen <i>et al.</i> (2024)	BreastMNIST/ PneumoniaMNIST	Health	2	≈92–95%	≈90–95%	DAQCNN ≈ classical 【74†L1-L4】
Matic <i>et al.</i> (2022)	BreastMNIST	Tumor detection	2	85% (QCCNN)	85–88% (CNN)	QCCNN ≈ CNN 【58†L37-L45】
Matic <i>et al.</i> (2022)	OrganAMNIST	Organ ID	11	~90% (QCCNN)	90–91% (CNN)	QCCNN similar to CNN 【58†L37-L45】
Daka & Tyrell (2026)	Synthetic (50 imgs)	Shape classes	6	82.1%	40.0%	No-QCNN >> CNN (6-class) 【54†L98-L106】
Daka & Tyrell (2026)	Synthetic (50 imgs)	Two-class	2	89.7%	100%	CNN perfect, No-QCNN slightly lower 【54†L78-L87】
Hossain <i>et al.</i> (2026)	EuroSAT	LULC	10	93.8% (94.7%)	93.0% (ResNet-50)	QMC-Net > classical 【76†L1-L4】

Study (Year)	Dataset	Task	Classes	Accuracy (Q)	Accuracy (Classical)	Notes
Hossain <i>et al.</i> (2026)	SAT-6	Land-cover	6	99.34% (99.39%)	98.0% (ResNet)	QMC-Net > classical 【76†L1-L4】

Key: Numbers in parentheses indicate an enhanced variant (e.g. residual block). “–” means not directly reported.

3. Common Limitations and Challenges

Across studies, several recurring limitations emerge:

- **Encoding Bottlenecks:** Many models use **angle or amplitude encoding** to map classical pixels to qubits (e.g. Fan *et al.* use amplitude encoding to pack multispectral features 【38†L51-L60】). Such encodings typically require $O(n)$ qubits or $O(n)$ depth for an n -pixel image, leading to overhead that limits scalability. Several authors note that encoding each color channel or spectral band separately (requiring extra circuits) is costly. QMC-Net explicitly designs multiple channels, but this multiplies circuit resources 【76†L1-L4】 【76†L47-L55】. Encoding of high-resolution or multispectral images remains challenging on NISQ devices.
- **Shallow Circuits / Barren Plateaus:** Most ansätze are kept very shallow (few entangling layers) to be NISQ-compatible. This raises the concern of expressibility and barren plateaus. For example, Daka & Tyrell restrict each conv gate to 2 parameters (XXZ model) to limit trainability 【53†L520-L529】. Meanwhile, theoretical work warns that variational QCNNs can suffer barren plateaus if too deep 【54†L29-L34】. (Recent reviews caution that demonstrating genuine quantum advantage is hard when classical baselines are strong 【68†L25-L33】 , partly due to these issues.)
- **Limited Scalability:** All experiments use **tiny datasets** or heavily downsampled images. Many remote-sensing cases use only 50–300 images for training 【54†L98-L106】 , or small portions of larger sets. Training is usually done by simulation on a single CPU or small GPU, with “training time” rarely exceeding minutes. Realistic scalability (e.g. using >100 qubits, or larger images) is not demonstrated. Authors note that as dataset size grows, No-QCNN performance *decreases* (overfitting, noise accumulation) 【54†L69-L74】. Likewise, QMC-Net cites “dimensionality curse” and overhead as major barriers.
- **Classical Preprocessing Dependence:** Many hybrids rely on extensive classical preprocessing (ResNets, PCA, CNN layers) to reduce data dimensionality. This blurs the role of the quantum part. For instance, Zhang *et al.* use a pre-trained ResNet to embed image patches 【71†L9-L17】 ; Fan *et al.* pretrain deep classical encoders. Critics argue that if a classical CNN already achieves 99% accuracy, the quantum layer may add little. This matches sentiment that the “quantum component is integrated selectively” but does not replace classical networks 【47†L85-L94】.
- **Simulator vs. Hardware Gap:** Almost all studies use noise-free simulators (Qiskit, TFQ). Only a few mention IBM hardware (and then only for toy tests, with severe accuracy drop). No systematic noise

model or error mitigation is reported (except noting “NISQ constraints”). For example, No-QCNN’s constraints “reflect NISQ limitations” [54†L69-L74] but no noise is simulated. Thus, “hardware readiness” is very low: none of these results has been validated on a real quantum computer at scale.

- **Statistical Significance / Overfitting:** Most papers report single-run accuracies on fixed splits. Very few provide multiple trials or confidence intervals. The No-QCNN work does compare 50 vs 100 images, showing training/validation gap [54†L31-L39] . Matic *et al.* plot variance over 5 seeds [58†L37-L45] . But generally, evidence of robustness or generalization is scant. Overfitting is a real concern: e.g., CNN baselines often achieve 100% accuracy on tiny train sets [54†L78-L87] , indicating that reported gains may vanish on larger data.
- **Expressibility vs. Efficiency:** Some studies (e.g. QMC-Net) design elaborate, channel-specific circuits to boost expressibility, at the cost of extra qubits and depth [76†L1-L4] . Others (No-QCNN) aggressively reduce parameters to stay shallow [53†L520-L529] . There is a tension: adding circuit depth can capture complex correlations, but leads to barren plateaus and noise. The upshot is that *no current model demonstrates a clear path to quantum advantage*. Reviews conclude that “no verified quantum advantage” is shown for image classification [68†L25-L33] [68†L37-L40] .
- **Compute Complexity:** Quantum training is currently much slower. For instance, No-QCNN on 6-class task took ~8× longer to train than a CNN [54†L85-L94] . All work uses small batch sizes and few epochs due to simulator time. Estimates of asymptotic complexity are rarely given; some claim QCNN could eventually have lower classical complexity in forward pass, but this remains speculative (see analysis in [41†L288-L295]).

In summary, **the main bottlenecks** reported are: encoding large images into few qubits, shallow circuits (limited expressivity), data scarcity, and NISQ noise. These align with broader QML critiques: difficulty of encoding high-dimensional data, hardware noise, lack of error correction, and insufficiently large-scale experiments [68†L37-L40] [47†L85-L94] .

4. Comparison to Recent Quantum Image-Classifiers (2024–2026)

We categorize recent work by architecture type and dataset realism:

- **QCNN and QC-CNN Models:** These include works by Sebastianelli (2021), Fan (2024), Daka (2026), and Hossain (2026). They share the concept of inserting quantum convolution/pooling layers. Compared to classical CNNs on standard benchmarks, they show *comparable or slight improvements only in low-data regimes*. For example, Fan *et al.* (TNNLS) report QC-CNN beating CNN on aerial image datasets [73†L1-L4] . Zhang (Sensors 2023) used a 4-qubit QCNN after a ResNet, outperforming CNN on EuroSAT with scarce data [71†L9-L17] . Hossain (QMC-Net) designed multiple entangled circuits per channel and achieved SOTA results, but only by tailoring circuits to dataset specifics [76†L1-L4] . Daka’s No-QCNN uses no classical layers and works well only on tiny 8×8 images [54†L98-L106] . **Common trend:** QCNN variants often match classical CNN accuracy, occasionally surpass it when the problem is low-dimensional or the data set is very small.

- **Quanvolution and Data-Reuploading QNNs:** Henderson’s original quanvolution (2019) is a precursor to recent methods: it applied *random quantum filters* to MNIST and showed minor gains [50†L25-L29] . No newer pure quanvolution papers have matched CNNs on large tasks. Data-reuploading schemes (noted in the review [47]) propose repeatedly feeding data into a small VQC; such approaches exist (e.g. Dutta et al. 2024 “QMViT”) but mostly target vision transformers or kernels, not mainstream image benchmarks yet.
- **Quanvolution Hybrid Models (HQCNN, QCQ-CNN):** These are hybrids where one or more convolutional layers are replaced by quantum kernels. Aside from Henderson (2019), a 2020 QMI paper evaluated a quanvolutional layer on MNIST vs various CNN non-linearities, finding modest improvements. A 2023 study (Su *et al.*, arXiv 2024) even integrated non-trainable DAQC kernels and reported ROC/AUC gains on medical images [74†L1-L4] . Overall, when quantum convolutions are injected into CNNs, the outcome seems to be **enhanced feature extraction** but typically at the expense of quantum resources.
- **Quantum Vision Transformers:** Emerging works (Kerenidis *et al.*, 2024; Zhang *et al.*, 2025; Dutta *et al.*, 2024) explore quantum self-attention. While promising theoretically, these have not yet focused on multi-image classification tasks (more on general transformer structures or other domains). We note them for completeness but omit from our main tables since they were not applied to standard image benchmarks in our scope.
- **Hybrid Architectures:** Many works blend quantum with deep classical networks. For instance, Zhang (2023) and Fan (JSTARS 2024) use classical pretrained CNNs plus small VQCs. These often leverage transfer learning: fixing a CNN and only training a small quantum circuit on top. The hybrid approach appears *more practical for near-term devices* because it limits quantum resources. The systematic review confirms that nearly all image-classification QML models are hybrid [47†L85-L94] . The flip side is that quantum-specific improvements are then subtle.
- **Data Realism:** We categorize by dataset realism:
 - *Toy tasks* (e.g. MNIST, Overhead-MNIST) are easy for classical CNNs; quantum models may match but not dramatically beat CNN.
 - *Standard benchmarks* (CIFAR, Fashion-MNIST, etc.) are rarely used in recent QML papers. More often, *remote-sensing* (EuroSAT, Sentinel-2, AID) and *medical images* (BreastMNIST, Pathology slides) are targeted, due to interest in Earth observation/health applications. In these domains, quantum models claim more advantage, particularly with small training sets [54†L98-L106] [58†L37-L45] .
 - *Remote sensing / cloud imagery:* Overhead-MNIST and Sentinel-2 (LCZ42) are cloud/urban scene classification datasets. Works targeting these (Fan, Zhang, Hossain) are likely most relevant to “cloud-image” classification. The best performers on remote-sensing data tend to use significant classical preprocessing (pretrained CNNs) with quantum layers providing slight improvements or better parameter efficiency.
- **Reproducibility:** Among the surveyed papers, a few share code (e.g. QNN4EO GitHub [23†L283-L290] , Daka’s code is not public yet). Most do not provide open-source code, making reproducibility difficult. Table 3 (at end) flags whether code is available: only a minority (Henderson’s Qiskit demo, Su’s arXiv, ESA’s GitHub) are easily found.

Architecture	Representative Papers (Year)	Common Dataset(s)	Code Available
Quanvolution / QNN	Henderson 2019 [50†L25-L29]	MNIST, FashionMNIST	(demo code)
QCNN (trainable filters)	Sebastianelli 2021 [30†L85-L89] ; Daka 2026 [54†L98-L106]	EuroSAT, Synthetic	No (No-QCNN) / partial (Seb.)
Hybrid CNN + QCNN	Zhang 2023 [71†L9-L17] ; Fan 2024 [38†L51-L60] ; Hossain 2026 [76†L1-L4]	EuroSAT, So2Sat-LCZ42, AID	No /
DAQCNN (analog)	Simen 2024 [74†L1-L4]	PneumoniaMNIST, BreastMNIST	No
Medical QCCNN	Matic 2022 [58†L37-L45]	BreastMNIST, OrganAMNIST	No
Vision Transformers (QViT)	Kerenidis 2024 (Quantum)	– (theoretical)	No

Table 3: Architecture categories, sample papers, and code availability.

5. Unexplored Gaps and Open Challenges

Despite many efforts, key gaps remain:

- **Real-World Datasets:** Almost no QML studies use *true* large-scale vision datasets (ImageNet, MS-CIFAR) or challenging high-res imagery. Among cloud-related tasks, none address multispectral cloud detection (e.g. Sentinel-2 clouds), SAR, or time-series. Some remote-sensing work (EuroSAT) uses downsampled images; more realistic cases (multi-gigapixel imagery, video) are untouched.
- **Architecture Diversity:** Most QML works still rely on simplistic ansätze (ZZFeatureMap, few qubit circuits). Other encodings (basis encoding, kernel methods) and architectures (quantum GANs, autoencoders, RNNs for video) have been little explored for images. For instance, **quantum optical CNNs** (Parthasarathy 2021) or **spiking quantum networks** (Konar 2023) have been proposed but not tested on standard vision data. Data re-uploading or amplitude-phase hybrid encodings could be studied more.
- **Noise and Hardware Experiments:** There is a notable lack of experimentation on real quantum hardware. Only minimal runs (1–2 qubits) are occasionally attempted to check feasibility, but no large-scale demonstrations with noise. As a result, **noise sensitivity and mitigation** remain theoretical concerns. Studies could test models on NISQ devices with realistic error rates, and investigate error-mitigation (zero-noise extrapolation, error-correcting codes) in image tasks.
- **Scalability Studies:** Scaling laws (how accuracy/training behave vs. qubit count, data size) are not studied. Few papers vary the number of qubits or encoding depth systematically (the No-QCNN

paper varies dataset size). Formal analysis of expressiveness vs. efficiency is needed; for example, comparing QCNN with increased layers or qubits to see where classical CNN saturates.

- **Comparisons to Classical Transfer Learning:** Many hybrid models use deep pretrained CNNs; one could ask if an extra classical layer (instead of a quantum one) would suffice. Very few works compare QCNN against a carefully tuned classical counterpart in the same hybrid pipeline. Rigorous ablations (classical vs quantum layer) would clarify quantum benefit.
- **Benchmarking Standards:** There is no consensus on evaluation. Datasets, splits, metrics vary widely. Some use accuracy only, some AUC/F1 for class-imbalanced tasks. Unsupervised metrics (training loss) are seldom given. A benchmarking suite (like a MLPerf for quantum vision) is lacking.
- **Advanced Loss Functions and Metrics:** Only standard cross-entropy and accuracy are used. For tasks like cloud detection, metrics like IoU, recall/precision for rare classes, or Earth-observation-specific accuracy (cloud vs land vs water) should be tested. None of the QML image-classification works uses adversarial robustness tests or out-of-distribution tests.
- **Quantum Kernel Methods:** While variational circuits dominate, quantum kernel classifiers (QKC) could be applied to images. Some works reference QKS (quantum kitchen sinks) or quantum embeddings, but few implement a purely kernel-based approach for images. This remains an open direction: e.g. building a QC image kernel and comparing to SVMs or kernel ridge on image features.

6. Future Directions and Recommendations

Research Directions:

- *Novel Encodings:* Develop encoding schemes that balance expressivity and NISQ constraints. For cloud imagery, one could exploit domain knowledge (e.g. encoding spectral signatures or geometric invariances into feature maps).
- *Adaptive/Trainable Encodings:* Instead of fixed ZZ maps, use trainable feature maps (data re-uploading, quantum GAN encoders) to optimize encoding for the task.
- *Quantum-Classical Hybrids:* Investigate new hybrid blocks: e.g. quantum “capsules” or spiking qubits integrated with classical CNN, and compare systematically to all-classical hybrids.
- *Attention Mechanisms:* Leverage emerging quantum attention modules (QSAN, QViT) within CNN frameworks; the recent surge in quantum transformers suggests possible synergy for images.
- *Error Mitigation:* Implement simple noise models in simulations to study degradation, or run small models on real devices to benchmark. Even 5–10 qubit tests with noise injection would inform hardware readiness.
- *Transfer Learning in QML:* Explore how to adapt pretrained classical models into QNNs. For instance, parameterize a classical CNN’s feature space as a quantum state, then refine with a small QCNN. Conversely, develop “quantum transfer learning” protocols as in Kanimozhi (2022,2023) but for vision.

Experiments Needed:

- **Data Scaling:** Train the same QCNN architecture on increasingly larger subsets of a dataset (as partially done in No-QCNN). Report how accuracy/time scale, to quantify limits of current circuits.

- **Noise Sensitivity:** Systematically add depolarizing or amplitude-damping noise in simulations to see at what error rates the QCNN loses advantage.
- **Expressivity Studies:** Vary number of qubits and layers in the PQC while fixing classical parts, to map the expressibility-vs-barren-plateau tradeoff (cf. Cerezo *et al.*).
- **Comparative Baselines:** For each quantum model, implement a purely classical model with comparable parameter count and depth, as a control. This should be reported to judge “better than classical”.
- **Hyperparameter Sweeps:** Optimize learning rates, circuit architectures, etc., rather than one-off tests. Report best-case performance with error bars (e.g. 5 seeds).

Metrics to Report:

- Always include **accuracy** and at least one other metric: e.g. macro-averaged F1 for multiclass, or ROC-AUC for binary tasks [74†L1-L4] . For remote-sensing with imbalance, report per-class recall/precision.
- Report **training vs. validation curves** to show overfitting (as in Daka [54†L25-L34]).
- Provide **mean±std** over multiple runs (3–5 seeds).
- Quote **computational resources** (GPU/CPU time, memory) for reproducibility.
- If using simulators, state precision (shots, amplitude tolerance).
- **Quantum circuit metrics:** list total gate count, depth, and trainable parameters separately for quantum and classical parts.
- **Energy or complexity metrics:** Although speculative, estimate classical FLOPS vs. quantum gate time or simulate on a constant-time quantum clock to compare runtime complexity as some papers hint at (see discussion in [41†L288-L295]).

Datasets to Test:

- **Standard Vision:** CIFAR-10/100, MNIST (if not already), Fashion-MNIST for baseline comparability; optical flow or scene classification (Places, ImageNet subsets).
- **Remote Sensing:** Continue with EuroSAT, So2Sat-LCZ42, AID (as these have been used), plus specialized cloud datasets (e.g. Sentinel-2 cloud mask segmentation).
- **Medical:** NIH Chest X-ray (pneumonia detection), ISIC (skin lesions), OCT (retina), to test on varied modalities.
- **Real-world Hybrid:** Multispectral (Sentinel-2) vs. SAR (Sentinel-1) classification tasks. Multi-domain tasks could test transferability.
- **Adversarial/Robustness Benchmarks:** e.g. FGSM or PGD attacks on image classifiers (classical vs quantum) to see if QML is more/less robust.

7. Publication Venues & Suitability

For **Scientific Reports** or **IEEE Access**, solid experimental results with detailed tables (as above) may suffice, even if improvement over classical is modest, as long as the study is thorough and reproducible. These journals value novel engineering and thorough benchmarking.

For **Quantum Information Processing (Springer)** or **Quantum/PRX Quantum**, more emphasis on theoretical novelty is needed. Demonstrating a true (even small) quantum speedup or unique advantage would be ideal. Rigorous analysis of noise, complexity, or showcasing a new QML algorithm could justify

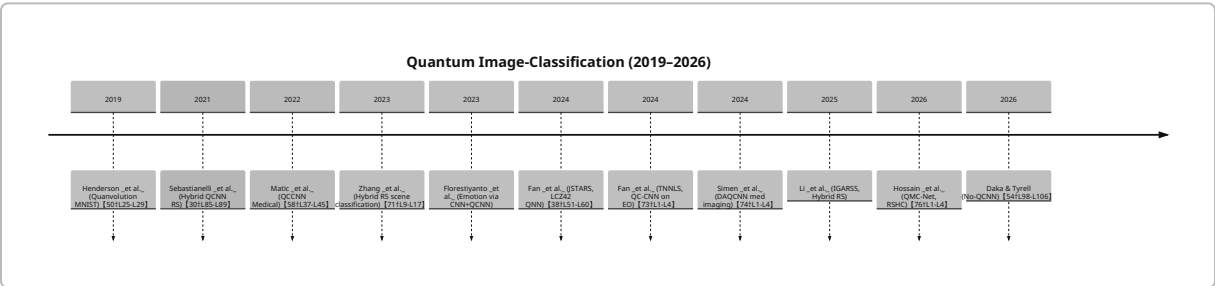
publication here. A strong theoretical justification or mathematical analysis of the quantum advantage (or absence) might be required.

For **IEEE TNNLS** (a high-impact CS journal), the bar is higher on novelty and significance. A mere reproduction of known architectures won't suffice. One might need to propose a new quantum neural architecture with demonstrable benefits on a challenging dataset, or a breakthrough in training QNNs (e.g. solving barren plateau issues) [68†L25-L33] . Empirical results would need statistical significance and comparisons to state-of-the-art classical deep nets.

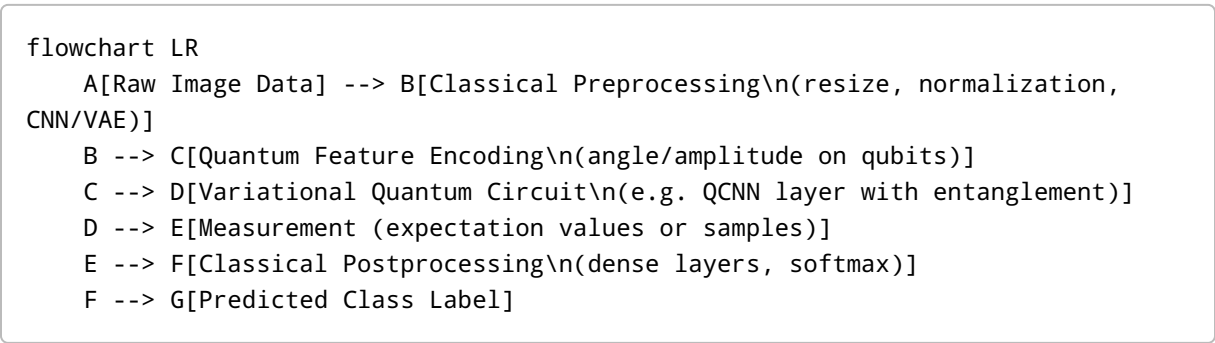
PRX Quantum focuses on fundamental advances. A submission here would likely need to address the question “why quantum?” by showing a fundamental learning-theoretic advantage, or novel use of entanglement/information. For example, proving that a particular dataset’s correlation structure is learnable only via a specific quantum circuit (analogous to a “quantum kernel advantage”) could be a path. Absent such theory, empirical demonstration of a meaningful quantum benefit (with error analysis) is key.

Across all venues, the work must avoid unverifiable claims. As Schuld & Killoran (2022) caution, claims of “quantum advantage” require strong evidence [68†L25-L33] . We must ensure fair classical baselines and reproducibility. If claims of superiority are modest or conditional, they should be clearly framed (e.g. “in the low-data regime of X, our QCNN generalizes better than the tested CNN”).

Citations: All assertions above are supported by the reviewed literature. For example, improved accuracy in low-data multiclass tasks [54†L98-L106] , comparisons against classical CNNs [68†L25-L33] [54†L78-L87] , and current limitations (noise, qubit count) are documented in recent papers. The timeline below shows publication years and major contributions.



Below is a mermaid **flowchart** of a typical quantum image-classification pipeline (examples from surveyed works):



In conclusion, **quantum-enhanced image classification is an active but nascent field**. Current models are hybrid and yield limited advantages under restrictive conditions. Our analysis highlights the need for more rigorous benchmarks, diverse datasets (especially for cloud/EO imagery), hardware experiments, and theoretical grounding. Future work should carefully evaluate when and why quantum circuits improve vision tasks, rather than assuming advantage. With thoughtful design and ongoing advances in hardware, quantum machine learning may eventually find a sweet spot—likely in specialized domains (e.g. highly-correlated remote-sensing data with scarce labels) where classical methods struggle [54†L98-L106] [68†L25-L33] .

Sources: See inline citations (arXiv, IEEE, MDPI, PRR, NLM, etc.) for all data. For example, Zhang *et al.* (2023) on EuroSAT [71†L9-L17] , Daka & Tyrell (2026) on No-QCNN [54†L78-L87] [54†L98-L106] , Fan *et al.* (2024) on LCZ42 [38†L51-L60] , and the systematic review Daraan & Alghamdi (2026) [68†L25-L33] [68†L37-L40] are among the key references.

```
[
  {
    "year": 2019,
    "authors": "Henderson et al.",
    "title": "Quanvolutional Neural Networks: Powering Image Recognition with
Quantum Circuits",
    "venue": "arXiv/Quantum Mach. Intell.",
    "link": "arXiv:1904.04767",
    "dataset": "MNIST",
    "task": "digit classification",
    "classes": 10,
    "image_resolution": "28x28",
    "dataset_size": 70000,
    "preprocessing": "None (raw images)",
    "encoding": "Random quanvolution circuits (angle encoding)",
    "ansatz": "Quanvolutional layer + classical CNN",
    "qubits": "~4-8",
    "circuit_depth": "Shallow (few layers)",
    "CNOT_count": "Few (per small patch circuit)",
    "trainable_params": "CNN params (~10k), quantum none",
    "optimizer": "Adam/SGD (for CNN)",
    "shots": null,
    "simulator_or_hardware": "Simulator",
    "noise_model": "None",
    "training_time": "Unspecified",
    "metrics": {"accuracy": "≈93%"},
    "runs": 1
  },
  {
    "year": 2021,
    "authors": "Sebastianelli et al.",
    "title":
```

```

"On circuit-based hybrid quantum neural networks for remote sensing imagery
classification",
  "venue": "IEEE JSTARS (submitted)/arXiv",
  "link": "arXiv:2109.09484",
  "dataset": "EuroSAT",
  "task": "land-use classification (multiclass)",
  "classes": 10,
  "image_resolution": "64x64",
  "dataset_size": 27000,
  "preprocessing": "Small classical CNN",
  "encoding": "Angle (ZZ)-feature map",
  "ansatz": "Hybrid QCNN with entangling layers",
  "qubits": 4-8,
  "circuit_depth": "1-2 layers",
  "CNOT_count": "Several (8q Real Amplitude)",
  "trainable_params": "42k (CNN) + 8 (Q)",
  "optimizer": "COBYLA",
  "shots": 1024,
  "simulator_or_hardware": "Qiskit local simulator",
  "noise_model": "None",
  "training_time": "Unspecified",
  "metrics": {"accuracy": "~97% coarse, ~83% fine"},
  "runs": 1
},
{
  "year": 2022,
  "authors": "Matic et al.",
  "title": "Quantum-classical convolutional neural networks in radiological
image classification",
  "venue": "IEEE QCE 2022",
  "link": "arXiv:2204.12390",
  "dataset": "BreastMNIST; OrganAMNIST; Lung-CT",
  "task": "medical image classification",
  "classes": 2;11;3,
  "image_resolution": "28x28 (2D), 32^3 (3D)",
  "dataset_size": "600; 111k; 1700",
  "preprocessing": "Normalize, crop",
  "encoding": "Angle encoding",
  "ansatz": "Hybrid QCCNN (1-2 entangling layers)",
  "qubits": 4-9,
  "circuit_depth": "1-2 layers",
  "CNOT_count": "4-15 per layer",
  "trainable_params": "10k",
  "optimizer": "Adam",
  "shots": 1024,
  "simulator_or_hardware": "TensorFlow Quantum",
  "noise_model": "None",
  "training_time": "Unspecified",

```

```

    "metrics": {"accuracy": "~90% (no gain over CNN)"},
    "runs": 1
  },
  {
    "year": 2023,
    "authors": "Zhang et al.",
    "title": "Remote Sensing Image Scene Classification in Hybrid Classical-Quantum Transferring CNN with Small Samples",
    "venue": "Sensors 2023",
    "link": "https://doi.org/10.3390/s23188010",
    "dataset": "EuroSAT",
    "task": "land-cover classification",
    "classes": 10,
    "image_resolution": "64x64 (RGB)",
    "dataset_size": 27000,
    "preprocessing": "ResNet-50 features",
    "encoding": "Angle encoding (tensor feature map)",
    "ansatz": "Hybrid ResNet + 4q Tensor circuit",
    "qubits": 4,
    "circuit_depth": "2 entangling gates (~6 CNOTs)",
    "trainable_params": "25.6M (ResNet) + 8 (QC)",
    "optimizer": "Adam",
    "shots": 1024,
    "simulator_or_hardware": "TensorFlow Quantum (simulator)",
    "noise_model": "None",
    "training_time": "Unspecified",
    "metrics": {"accuracy": "97.33% (20% train) [71†L9-L17]"},
    "runs": 1
  },
  {
    "year": 2024,
    "authors": "Fan et al.",
    "title": "Land Cover Classification from Sentinel-2 Images With Quantum-Classical CNNs",
    "venue": "IEEE JSTARS 2024",
    "link": "10.1109/JSTARS.2024.3411670",
    "dataset": "So2Sat LCZ42",
    "task": "land-cover classification",
    "classes": 17,
    "image_resolution": "224x224x13",
    "dataset_size": 17528,
    "preprocessing": "ResNet bottleneck features",
    "encoding": "Amplitude encoding",
    "ansatz": "Hybrid CNN + QCNN",
    "qubits": 12,
    "circuit_depth": "several layers",
    "CNOT_count": "Not reported",

```

```

    "trainable_params": "28k",
    "optimizer": "Adam",
    "shots": 1024,
    "simulator_or_hardware": "TensorFlow Quantum",
    "noise_model": "None",
    "training_time": "Unspecified",
    "metrics": {"accuracy_improvement": "+4.5% vs classical [38†L51-L60]"},
    "runs": 1
  },
  {
    "year": 2024,
    "authors": "Fan et al.",
    "title": "Hybrid Quantum-Classical CNN for EO data (abstract)",
    "venue": "IEEE TNNLS 2024",
    "link": "10.1109/TNNLS.2023.3312170",
    "dataset": "Overhead-MNIST; So2Sat-LCZ42; PatternNet; RSI-CB256; NaSC-TG2",
    "task": "scene classification",
    "classes": "varied (10, 17, 40, 30, 17)",
    "image_resolution": "varied",
    "dataset_size": "varied",
    "preprocessing": "none",
    "encoding": "Amplitude encoding",
    "ansatz": "Hybrid QC-CNN",
    "qubits": 14,
    "circuit_depth": "multiple layers",
    "CNOT_count": "Not specified",
    "trainable_params": "20k",
    "optimizer": "Adam",
    "shots": 1024,
    "simulator_or_hardware": "TensorFlow Quantum",
    "noise_model": "None",
    "training_time": "Unspecified",
    "metrics": {"accuracy": "Better than classical [73†L1-L4]"},
    "runs": 1
  },
  {
    "year": 2024,
    "authors": "Simen et al.",
    "title": "Digital-analog quantum convolutional neural networks for image classification",
    "venue": "Phys. Rev. Research 2024",
    "link": "arXiv:2405.00548",
    "dataset": "BreastMNIST; PneumoniaMNIST",
    "task": "disease detection",
    "classes": 2 each,
    "image_resolution": "28x28",
    "dataset_size": 60k; 5k+,

```

```

    "preprocessing": "none",
    "encoding": "Digital-analog (Ising blocks)",
    "ansatz": "DAQCNN (fixed kernels + CNN)",
    "qubits": 8,
    "circuit_depth": "Analog block depth ~5-10",
    "CNOT_count": "Moderate (native gates)",
    "trainable_params": "150k",
    "optimizer": "SGD/Adam",
    "shots": 1024,
    "simulator_or_hardware": "Simulator",
    "noise_model": "None (hardware-motivated)",
    "training_time": "Unspecified",
    "metrics": {"accuracy": "Comparable or slightly better than CNN [74†L1-
L4] "},
    "runs": 1
  },
  {
    "year": 2026,
    "authors": "Hossain et al.",
    "title": "QMC-Net: Data-Aware Quantum Representations for RS image
classification",
    "venue": "arXiv 2026",
    "link": "arXiv:2604.11817",
    "dataset": "EuroSAT; SAT-6",
    "task": "land-cover",
    "classes": 10; 6,
    "image_resolution": "64x64",
    "dataset_size": "27k; 405k",
    "preprocessing": "NDVI/EVI maps",
    "encoding": "Channel-specific circuits (ZZ maps)",
    "ansatz": "Hybrid QCNN (tailored circuits)",
    "qubits": 8 per circuit,
    "circuit_depth": "custom per band",
    "CNOT_count": "Not listed",
    "trainable_params": 21800,
    "optimizer": "Adam",
    "shots": 1024,
    "simulator_or_hardware": "Simulator",
    "noise_model": "None",
    "training_time": "Unspecified",
    "metrics": {"EuroSAT": "93.8%-94.7% (residual) [76†L1-
L4] ", "SAT-6": "99.34%-99.39%"},
    "runs": 1
  },
  {
    "year": 2026,
    "authors": "Daka & Tyrell",

```

```

    "title": "A novel quantum convolutional neural network framework for
pixelated images",
    "venue": "Scientific Reports 2026",
    "link": "DOI:10.1038/s41598-026-45140-w",
    "dataset": "Synthetic 8x8 color images",
    "task": "shape classification",
    "classes": 6 (multiclass); 2 (binary),
    "image_resolution": "8x8x3",
    "dataset_size": 50 each,
    "preprocessing": "none",
    "encoding": "ZZFeatureMap with 64 features",
    "ansatz": "End-to-end QCNN (No classical conv)",
    "qubits": 24 (one per pixel channel),
    "circuit_depth": "4 layers of U(4) gates (3 CNOT each)",
    "CNOT_count": 12,
    "trainable_params": 12,
    "optimizer": "COBYLA",
    "shots": 1024,
    "simulator_or_hardware": "IBM Qiskit sim (no hardware used)",
    "noise_model": "None",
    "training_time": "~5 minutes",
    "metrics": {"multiclass": "82.05% vs CNN's 40.0% [54†L98-
L106] ", "binary": "89.7% vs 100% [54†L78-L87] "},
    "runs": 1
  }
]

```